FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE II: FREE ELECTRON THEORY AND BAND THEORY

Past questions

Section A: 1 to 3 marks

Drude–Lorentz classical theory

1 What is the limitation of free electron model in explaining Hall coefficients of metals ? **(1 weightage)** *November 2019 (M.Sc.)*

2 Find the relaxation time and mean free path of conduction electron in copper. For copper density of the free electron $8.5 \times 10^{28}/m^3$, resistivity 1.69×10^{-8} ohm-m and average velocity of electron 1.154×10^5 m/s. **(3 weightage)** *November 2021 (M.Sc.)*

3 What is Wiedemann -Franz law ? **(1 weightage)** *November 2021 (M.Sc.)*

4 What is Hall effect ? Give the expression for Hall co-efficient as predicted by free electron theory. **(1 weightage)** *November 2021 (M.Sc.)*

5 Briefly explain Hall effect. **(1 weightage)** *November 2023 (M.Sc.)*

Free electron gas: Fermi energy, density of states and applications

6 Valence electron density for monovalent potassium is, $N/V = 1.402 \times 10^{28}\ m^{-3}$. For a typical current density of $10^6\ Am^{-2}$, obtain the drift velocity in terms of Fermi velocity for electrons in potassium. **(3 weightage)** *November 2019 (M.Sc.)*

7 Find the Fermi velocity of the electrons if the number density of electrons in sodium is $2.52 \times 10^{28}\ m^{-3}$ at room temperature. **(3 weightage)** *November 2022 (M.Sc.)*

8 The unit cell volume of sodium is $7.93 \times 10^{-29}\ m^3$. Calculate the Fermi energy of sodium at absolute zero **(3 weightage)** *November 2023 (M.Sc.)*

Band theory: Bloch theorem, Kronig–Penney model, velocity and effective mass

9 What is effective mass of a charge carrier ? What is its importance ? **(1 weightage)** *November 2019 (M.Sc.)*

10 What is effective mass of a charge carrier ? What is its importance ? **(1 weightage)** *November 2022 (M.Sc.)*

11 The energy near the valence band edge of a crystal is given by $E = -Ak^2$, where $A = 10^{-39}\ J\ m^2$. An electron with wave vector $k = 10^{10}\ k_x\ m^{-1}$ is removed from an orbital in the completely filled valence band. Determine the effective mass, velocity and momentum of the hole. **(3 weightage)** *November 2023 (M.Sc.)*

Metals, insulators and semiconductors

12 Explain the concept of holes in semiconductors. **(1 weightage)** *November 2019 (M.Sc.)*

13 Prove that the Fermi level in an intrinsic semiconductor lies in the middle of the forbidden gap if the effective masses of electrons and holes are equal. **(3 weightage)** *November 2019 (M.Sc.)*

14 What do you meant by indirect band gap semiconductors ? **(1 weightage)** *November 2023 (M.Sc.)*

Section B: 4 to 5 marks

Free electron gas: Fermi energy, density of states and applications

15 Explain the reason for a very low value of electronic heat capacity of metals at ordinary temperatures. **(5 marks)** *March 2023*

Band theory: Bloch theorem, Kronig–Penney model, velocity and effective mass

16 Discuss with necessary theory the Kronig-Penny model and show how energy bands are formed in solids. **(5 weightage)** *November 2021 (M.Sc.)*

17 Discuss the formation of allowed and forbidden energy bands on the basis of the Kronig Penny model. **(5 weightage)** *November 2023 (M.Sc.)*

Section C: 10 marks